Notes: Chari, Liu, Wang, & Wang (RES, 2021)

Property Rights, Land Misallocation, and Agricultural Efficiency in China

Contents

1 Big picture

- **Question.** Does formalizing farmers' right to lease land reduce land misallocation and raise agricultural productivity? The paper studies whether a property-rights reform in rural China improved the allocation of land across farmers and across crops.
- **Setting.** The Rural Land Contracting Law (RLCL), announced nationally in 2003, formally protected farmers' leasing rights. Provinces implemented the law at different times, creating staggered policy variation that the paper uses in a difference-in-differences design.
- **Core empirical object.** The reform did *not* give private ownership, sales rights, or mortgage rights. Its main innovation was to make land leasing contracts legally protected. That makes the paper really about the allocative value of *exchange rights*.
- **Main results.** After reform, land rented in rises by about 7% in the full sample and about 10% among farming households. At the village level, aggregate agricultural output rises by about 8% and aggregate revenue per mu by about 10%.
- **Mechanism.** Land is reallocated from low-productivity farmers to high-productivity farmers. Productive farmers cultivate more land and hire more labor after the reform, while less-productive farmers cultivate less land.
- **Most important quantitative decomposition.** About 88% of the increase in aggregate productivity comes from improved input reallocation rather than from an increase in average farmer productivity.
- **Dynamic allocative margin.** The reform also makes land allocation more responsive to crop-price movements. In other words, it improves not only static efficiency but also farmers' ability to adjust acreage across crops when relative prices change.
- **Why the paper matters.** Much of the misallocation literature is structural and asks what full removal of wedges would do. This paper gives quasi-experimental evidence from an actual policy change, so it is a rare causal estimate of how land-market frictions affect agricultural efficiency.
- **Main caveat.** The RLCL only partially reduced land distortions. Farmers still lacked full ownership and collateral rights, so the reform is best interpreted as a partial relaxation of one important friction rather than a full market liberalization.

2 Data and measurement (institutional context and key objects)

2.1 Institutional background and the reform

- **Pre-reform land system.** Under the Household Responsibility System, households had use rights over collectively owned land, but local governments retained substantial power over land assignments.
- **1998 reform versus 2003 reform.** The 1998 Land Management Law strengthened tenure security with 30-year contracts. The 2003 RLCL added legal protection for renting out and renting in land.

- **What changed economically.** Before 2003, informal leasing existed, often through verbal agreements among relatives and neighbors. The RLCL made those contracts legally enforceable and laid out procedures for disputes.
- **What did *not* change.** The RLCL did not create inheritance rights, mortgage rights, or full sales rights. That is why the paper isolates the effect of better leasing rights rather than broader privatization.
- **Why province-level implementation matters.** The central law explicitly allowed provinces to work out implementation details. The authors therefore date treatment using the timing of provincial implementation, which is the moment when the reform became operational in local practice.
- **Implementation window.** By the end of 2010, 22 provinces had officially implemented the RLCL.

2.2 Main household panel: National Fixed Point Survey (NFP)

- **Source.** National Fixed Point Survey run by the Research Center of Rural Economy under the Chinese Ministry of Agriculture.
- **Period.** 2003–2010. The authors start in 2003 because the survey structure changed substantially that year.
- **Coverage.** More than 19,000 households in 399 villages across 32 provinces, producing 157,315 household-year observations and 412,603 household-crop-year observations.
- **Why the data are valuable.** The panel is unusually large for an agricultural household survey and contains detailed crop-specific inputs and outputs.
- **Unit of observation.** Some outcomes are measured at the household-year level, some at the household-crop-year level, and some are aggregated to the village-year level.

2.3 Main variables and how they are constructed

- **Area cultivated.** Average household land cultivated is 12.37 mu, roughly 2 acres.
- **Land rented in.** Constructed as reported farmland minus land assigned by the village government. Average rented-in land is 2.227 mu, about 18% of cultivated area.
- **Land rented out.** Available only in 2009 and 2010, and separately recorded for land rented to individuals and to firms.
- **Household labor outcomes.** The survey records days spent in on-farm work, local off-farm work, and migrant work for each household member; the paper aggregates them to the household-year level.
- **Crop-level production inputs.** For each crop, the data record output quantity, area, labor days, machinery costs, and a bundle of other inputs such as fertilizer, pesticides, irrigation, and small tools.

- **Village-level aggregate output.** The authors aggregate crop outputs to the village level and value them at national crop-year prices. This removes local price variation, so estimated reform effects on revenue can be read as real output effects.
- **Transformations.** Many outcomes use the inverse hyperbolic sine IHS($\cdot$) transformation because the data contain many zeros and a long right tail.

2.4 Supplementary land quality and price data

- **Land quality survey.** In 2012 the authors surveyed a 10% subsample of NFP households. The matched sample contains 2,326 households in 247 villages across 20 provinces.
- **Land quality measure.** Households report plot quality as high, medium, or low; the authors average plot-level responses to build a household-level soil-quality control.
- **Chinese crop prices.** Provincial crop-price indices from *China Rural Statistics Yearbooks*, 2002–2009.
- **Global price instrument.** U.S. crop-price indices from the FAO over the same period. These are used as plausibly exogenous global price shifters, especially for cash crops.

2.5 Summary-statistic patterns that matter for the paper

- **Household scale.** Farms are very small on average, which makes within-village land misallocation quantitatively important.
- **Rental activity.** Rental markets exist pre-reform, but they are thin relative to the amount of land cultivated.
- **Labor allocation.** Over half of households have at least one migrant worker, and 42% report local off-farm work. This matters because one might expect better land markets to also change labor allocation.
- **Village productivity baseline.** Mean village-level aggregate revenue is about 1.03 million RMB and mean aggregate revenue per mu is 4,319 RMB.
- **Crop-level input use.** Mean crop area is 4.717 mu, mean machinery cost is 85.74 RMB, mean labor use is 68.10 days, and mean other input spending is 672 RMB.

3 Models and empirical specifications (all equations + interpretation)

3.1 Baseline household-level difference-in-differences

For household h in province p and year t, the main rental-activity regression is

$$y_{hpt} = \alpha + \beta_0 \,\mathrm{PostReform}_{pt} + \beta_1 \,\mathrm{ReformYear}_{pt} + \theta \,\mathrm{Tax}_{pt} + \gamma_t + \gamma_h + \varepsilon_{hpt}. \tag{1}$$

- **Dependent variable.** Usually IHS of land rented in, though the same structure is used for other household outcomes.

- **Why split reform year from post-reform years?** Many provincial laws were implemented late in the calendar year, often in October or November, so the reform year itself may contain only partial exposure.
- **Controls.** Household fixed effects, year fixed effects, and the provincial agricultural tax rate.
- **Interpretation of β_0.** The medium-run effect of implemented leasing rights on household rental behavior.

3.2 Event-study specification

To test for pre-trends and dynamic effects, the paper estimates

$$y_{hpt} = \alpha + \sum_{k=-3}^{2} \beta_k \,\text{Reform}_{pt,k} + \theta \,\text{Tax}_{pt} + \gamma_t + \gamma_h + \varepsilon_{hpt}, \tag{2}$$

where $\text{Reform}_{pt,k}$ indicates event time relative to the province's reform year and $k = -1$ is omitted.

- **Interpretation.** The event-study checks whether treated provinces were already on different trends before implementation.
- **Important design detail.** The sample includes provinces that had not implemented the reform by 2014, giving the event-study a pure control group.

3.3 Village-level output and productivity regression

For village v, province p, year t, the aggregate-outcome specification is

$$y_{vpt} = \alpha + \beta_0 \,\text{PostReform}_{pt} + \beta_1 \,\text{ReformYear}_{pt} + \theta \,\text{Tax}_{pt} + \gamma_t + \gamma_v + \varepsilon_{vpt}. \tag{3}$$

- **Dependent variables.** Log village aggregate revenue and log village aggregate revenue per mu.
- **Interpretation.** If leasing rights reduce land-market frictions, villages should experience higher output and higher land productivity once land shifts toward better farmers.

3.4 Crop production function used to construct productivity measures

At the household-crop-year level, the paper estimates a crop-specific Cobb–Douglas production function:

$$y_{icvt} = \alpha_c \log L_{icvt} + \beta_c \log N_{icvt} + \gamma_c \log K_{icvt} + \delta_c \log M_{icvt} + \phi_{icvt}, \tag{4}$$

where y_{icvt} is log physical output, L is land area, N labor days, K machinery costs, and M other input costs.

The unobserved productivity term is decomposed as

$$\phi_{icvt} = \phi_{ic} + \phi_{it} + \phi_{cvt} + e_{icvt}. \tag{5}$$

- **Interpretation of the fixed effects.** ϕ_{ic} is farmer-crop ability, ϕ_{it} is a farmer-year shock common across crops, and ϕ_{cvt} is a village-crop-year shock such as weather or pests.
- **Why this matters.** By including farmer-crop, farmer-year, and village-crop-year fixed effects, the paper tries to purge the production residual of the main components of omitted productivity heterogeneity that would otherwise bias input elasticities.
- **Main objects built from this step.** A household-crop fixed component of TFP, and a household-crop-year marginal product of land. Under Cobb–Douglas, the marginal product of land is proportional to average product times the land elasticity.
- **Average land elasticity.** Across crops, the average estimated output elasticity with respect to land is about 0.47.

3.5 Heterogeneous land reallocation by pre-reform productivity

To test whether land is reallocated toward more productive farmers, the paper estimates

$$\begin{aligned}\ell_{icvt} = \alpha + \sum_{j=1}^{4} \beta_j \psi_{icvt}^{j} + \sum_{j=1}^{4} \delta_j \left(\text{PostReform}_{pt} \times \psi_{icvt}^{j}\right) \\ + \sum_{j=1}^{4} \theta_j \left(\text{ReformYear}_{pt} \times \psi_{icvt}^{j}\right) + \eta_i + \eta_c + \eta_t + \varepsilon_{icvt},\end{aligned} \tag{6}$$

where ψ_{icvt}^{j} indicates the quartile of pre-reform marginal productivity.

- **Dependent variable.** Log crop area.
- **Key coefficients.** The δ_j terms show whether post-reform land growth is concentrated in high-productivity quartiles.
- **Economic prediction.** In a less distorted allocation, low-MPL farmers should shrink and high-MPL farmers should expand.

3.6 Olley–Pakes decomposition of aggregate productivity

Village-level aggregate productivity is written as

$$I_{vt} = \sum_{i,c} w_{ict}\phi_{ict} = \mathbb{E}(\phi_{ict} \mid v, t) + \sum_{i,c} \big(w_{ict} - \mathbb{E}(w_{ict} \mid v, t)\big)\big(\phi_{ict} - \mathbb{E}(\phi_{ict} \mid v, t)\big). \tag{7}$$

- **Interpretation.** Aggregate TFP is decomposed into average farmer-crop productivity plus a covariance term that measures whether large-output farmer-crops are also high-productivity farmer-crops.
- **Allocative meaning.** If resources are well allocated, the covariance term should be high because productive units command larger output shares.

The covariance term is further decomposed into within-crop and between-crop components:

$$
\begin{aligned}
\mathrm{Cov}^{\mathrm{OP}}_{vt} &\equiv \sum_{i,c} (w_{ict} - \mathbb{E}w_{ict})(\phi_{ict} - \mathbb{E}\phi_{ict}) \\
&= N\ \mathrm{cov}(w_{ict}, \phi_{ict}) \\
&= N\,\mathbb{E}[\mathrm{cov}(w_{ict}, \phi_{ict} \mid c)] + N\ \mathrm{cov}(\mathbb{E}[w_{ict} \mid c], \mathbb{E}[\phi_{ict} \mid c])\,.
\end{aligned}
\tag{8}
$$

- **Interpretation.** The paper asks whether the reform improves the matching of land and productivity within crops, across crops, or both.

3.7 Baily–Hulten–Campbell decomposition of productivity growth

Aggregate productivity growth is decomposed as

$$
\begin{aligned}
\Delta I_{vt} = &\sum_{i\in S} w_{ic,t-1}(\phi_{ict} - \phi_{ic,t-1}) + \sum_{i\in S} \phi_{ict}(w_{ict} - w_{ic,t-1}) \\
&+ \sum_{i\in E} w_{ict}\phi_{ict} - \sum_{i\in X} w_{ic,t-1}\phi_{ic,t-1},
\end{aligned}
\tag{9}
$$

where S denotes continuing farms, E entrants, and X exiting farms.

- **Interpretation.** Aggregate productivity growth can come from within-farm productivity growth, between-farm reallocation, entry, or exit.
- **Why it is useful here.** It directly checks whether the reform's gains come from better reallocation across existing farms rather than from technology shocks or sample turnover.

3.8 Benchmarking the gain from full reallocation

To compare the partial reform to a full-efficiency benchmark, the paper assumes

$$
Y_i = \phi_i L_i^{\alpha}, \tag{10}
$$

so aggregate village output can be written as

$$
Y = \left(\sum_i \phi_i s_i^{\alpha}\right) L^{\alpha}, \tag{11}
$$

where s_i is farmer i's land share and L is total land.

The efficient land share is

$$
s_i^* = \frac{\phi_i^{\frac{1}{1-\alpha}}}{\sum_j \phi_j^{\frac{1}{1-\alpha}}}. \tag{12}
$$

- **Interpretation.** Under the planner's allocation, more productive farmers should control proportionally more land.
- **Quantitative benchmark.** Comparing the actual land distribution to this efficient benchmark yields a mean output gain of about 73% from full reallocation.

3.9 Price-response design

To test whether the reform makes land allocation more responsive to crop-price movements, the paper estimates

$$\begin{aligned} Y_{hcvpt} = \alpha &+ \beta_0 (\text{Price}_{c,t-1} \times \text{PostReform}_{pt}) + \beta_1 (\text{Price}_{c,t-1} \times \text{ReformYear}_{pt}) \\ &+ \beta_2 \, \text{PostReform}_{pt} + \beta_3 \, \text{ReformYear}_{pt} + \nu_{ct} + \gamma_{hc} + \varepsilon_{hcvpt}. \end{aligned} \tag{13}$$

The dynamic version is

$$Y_{hcvpt} = \alpha + \sum_{k=-3}^{2} \left[\beta_k (\text{Price}_{c,t-1} \times \text{Reform}_{pt,k}) + \eta_k \, \text{Reform}_{pt,k} \right] + \nu_{ct} + \gamma_{hc} + \varepsilon_{hcvpt}. \tag{14}$$

- **Outcomes.** IHS crop area and an indicator for whether any land is allocated to the crop.
- **Why lagged prices?** Planting decisions are made before harvest, and using lagged prices also reduces simultaneity concerns.
- **Why U.S. prices?** U.S. crop prices are used as instruments for Chinese provincial crop prices, especially for cash crops whose local prices are tightly linked to global markets.

4 Identification and instruments (what makes the estimates credible)

4.1 Baseline identification idea

- **Treatment variation.** Provinces implemented the RLCL at different times after the national law was announced in 2003.
- **Comparison.** The design compares changes in outcomes in provinces after implementation to changes in provinces that have not yet implemented.
- **Key identifying assumption.** Absent the reform, treated and not-yet-treated provinces would have followed parallel trends in land rental activity, productivity, and other outcomes.

4.2 Main threats and the paper's responses

- **Endogenous timing of implementation.** The paper tests whether province-level adoption is predicted by rural income, agricultural employment, past land reallocations, land disputes, or pre-reform dispersion in land and marginal products. The answer is essentially no for the agricultural variables most relevant to the paper.
- **Pre-trends.** Event-study graphs for rental activity, aggregate output, aggregate productivity, land reallocation, and decomposition terms show little evidence of significant differential trends before reform.
- **Confounding policy changes.** The most important concurrent rural policy was the agricultural tax reduction. Every regression controls for the time-varying provincial agricultural tax rate, and results barely change.

- **Sample composition.** Results are stable in a balanced household panel, which helps address concerns that the findings are driven by sample entry or attrition.
- **Few-cluster inference.** For the sample restricted to adopting provinces, the paper uses Bertrand–Duflo–Mullainathan style randomization inference and finds similar significance patterns.

4.3 Checks specific to the land reallocation mechanism

- **Land quality.** The productive-farmer expansion result survives controls for household soil quality from the 2012 subsample survey.
- **Alternative productivity measures.** The same reallocation pattern appears when productivity is measured using fixed TFP components or agricultural profits per mu.
- **Production-function robustness.** Results are similar in balanced farmer-crop samples, with lagged-input instruments, and under a more flexible translog specification.

4.4 Checks on where the land goes

- **Renting out mirrors renting in.** In 2009–2010, the reform increases land rented out by about 7%, almost exactly matching the increase in rented-in land.
- **Not mainly to firms.** The increase in renting out is almost entirely to other individuals; renting out to firms is tiny and even slightly negative.
- **No disappearance of sampled land.** Total village land area under cultivation in the survey sample does not fall after the reform, which reduces the concern that land is moving mainly to unsampled entities.

4.5 Instrument for price-response regressions

- **First stage.** Chinese provincial cash-crop prices move closely with U.S. prices; the coefficient is 0.334 for cash crops and much smaller, 0.0908, for staple crops.
- **Interpretation.** U.S. prices are relevant instruments for Chinese cash-crop prices because global markets matter more there than for staple crops, whose domestic prices may be affected by state intervention.
- **Empirical consequence.** The price-response analysis therefore focuses on cash crops, where the exclusion restriction is most believable and the first stage is strongest.

5 Tables and figures

5.1 Table 1: Summary statistics

Read:

- Farms are tiny: mean cultivated area is only 12.37 mu, and mean rented-in land is 2.227 mu.

TABLE 1
Summary statistics

	Mean	Std Dev	Observations
Panel A: Variables by household-year			
Area (mu)	12.37	330.5	157,315
Land rented in (mu)	2.227	131.2	157,315
Land rented to individuals (mu)	0.329	2.433	41,577
Land rented to firms (mu)	0.0280	0.347	41,577
Total income	28,821.2	82,331.1	156,395
Any migration	0.552	0.497	156,441
Number of migrant work days	233.7	295.5	156,441
Any of-farm work	0.420	0.493	156,441
Number of of-farm work days	126.3	225.9	156,441
Total agricultural capital (RMB)	1,758.8	43,781.9	157,315
Panel B: Variables by village-year			
Aggregate revenue	1,031,183.9	21,739,192.6	2,486
Aggregate revenue per mu	4,319.0	155,214.2	2,485
Panel C: Variables by household-crop-year			
Output	6,242.9	1558191.8	412,603
Area (mu)	4.717	203.9	412,603
Machine inputs (RMB)	85.74	529.1	412,603
Labour inputs (days)	68.10	4,790.3	412,603
Other inputs (RMB)	672.0	74,541.3	412,603

Notes: Income and expenditures are in real 2002 renminbi. Panel B presents variables that we aggregate to the village level in the analysis. Panel C presents information on variables that are analysed at the crop level.

- The sample contains meaningful but still limited rental activity before the reform, which is exactly the situation where formal leasing rights can matter.
- Labor-market participation outside farming is common, but not universal, so one can meaningfully ask whether improved land markets trigger sectoral labor reallocation.
- The crop-level data are rich enough to estimate production functions and compute household-crop productivity measures.

5.2 Table 2 and Figure 1: Impact of reform on land renting

Read:

- In the full sample, post-reform land rented in rises by 0.0734, about 7%, significant at the 10% level.
- Among farming households the effect is larger, 0.0959, and in a balanced panel it is 0.110. This says the reform really affects active cultivators.
- In 2009–2010, total land rented out rises by 0.0726, almost all of it to individuals (0.0680), not firms (−0.0108).
- Figure 1 is the key parallel-trends check: there is no visible pre-trend in rented-in land, while coefficients turn positive immediately after reform.

5.3 Table 3 and Figure 2: Aggregate output and aggregate productivity

Read:

- Post-reform village aggregate revenue rises by 0.0792, about 8%.
- Revenue per mu rises by 0.104, about 10%, and is estimated more precisely.
- Figure 2 shows no meaningful pre-trend and a clear post-reform increase, especially for revenue per mu.

TABLE 2

Impact of reform on area rented in and rented out

	Renting in			Renting out		
	Full sample (1)	Restricted (2)	Balanced (3)	Total area (4)	Firms (5)	Individuals (6)
Post-reform year	0.0734*	0.0959**	0.110*	0.0726***	−0.0108	0.0680***
	(0.0394)	(0.0462)	(0.0548)	(0.0247)	(0.00657)	(0.0186)
Reform year	0.0713**	0.0795**	0.0693**	0.0577***	−0.00831**	0.0392***
	(0.0286)	(0.0315)	(0.0327)	(0.0123)	(0.00328)	(0.00929)
Observations	157,315	128,416	50,811	41,577	41,577	41,577

Notes: Each observation is a household-year. The dependent variables in Columns 1–3 are the IHS function of the area rented in, and IHS function of the area rented out in Columns 4–6. The sample in Column 1 includes all household-years. Column 2 restricts the sample to farming households. Column 3 further restricts the sample to households that were present in all periods. Columns 4 shows the total land rented out, Column 5 the area rented out to firms, and Column 6 the area rented out to individuals. The sample in Columns 4–6 includes only the 2009 and 2010 waves. The regressions include household fixed effects, year fixed effects and the agricultural tax rate. Standard errors clustered at province level are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

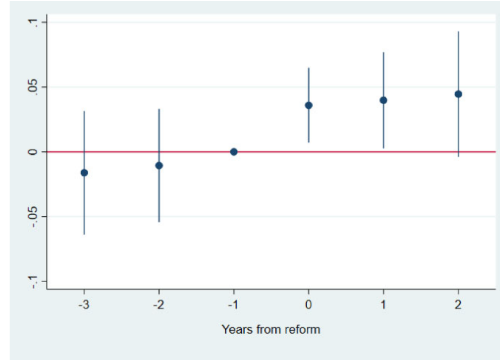


FIGURE 1

Leads and lags of reform effects on land renting

The figure plots the coefficients and associated 90% confidence intervals from estimating the leads and lags regression in equation 4.2, where the dependent variable is the IHS function of land rented in. All effects are relative to the year prior to the reform ($k = -1$).

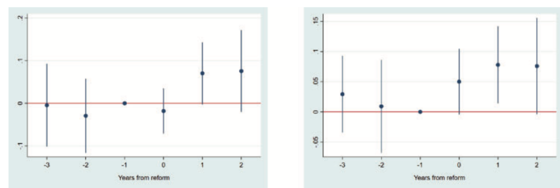
- These are village-level gains, so the reform is not just reshuffling observed land transactions; it is increasing real agricultural efficiency.

TABLE 3

Impact of reform on aggregate output and productivity

	Aggregate revenue (1)	Aggregate revenue per Mu (2)
Post-reform year	0.0792*	0.104***
	(0.0418)	(0.0290)
Reform year	−0.00562	0.0454
	(0.0370)	(0.0330)
Observations	2,232	2,232

Notes: The dependent variables are the logarithms of village-level aggregate revenue and aggregate revenue per mu respectively. The unit of observation is a village-year. All regressions include village fixed effects, year fixed effects, and the agricultural tax rate. Standard errors clustered at province level are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.



(a) Aggregate revenue **(b)** Aggregate revenue per mu

FIGURE 2

Leads and lags of reform effects on aggregate revenue and productivity

The panels in the figure plot the coefficients and associated 90% confidence intervals from estimating the leads and lags of reform effects, where the dependent variables are village-level agricultural revenue and village-level agricultural revenue per mu (both in logarithms). All effects are relative to the year prior to the reform ($k = -1$).

5.4 Figures 3, 4, and 5: Reallocation toward more productive farmers

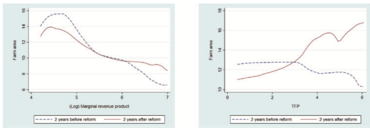
Read:

- Figure 3 gives the paper's most intuitive mechanism picture. Before the reform, farm area is weakly or perversely related to productivity; after the reform, high-MPL and high-TFP farmers operate more land.
- Figure 4 formalizes this: the post-reform effect on crop area is negative for the bottom productivity quartile and positive for the top quartiles, especially quartile 4.
- Figure 5 adds dynamics and shows that these reallocations happen after, not before, the reform.
- This is the cleanest evidence that aggregate productivity gains come from land moving toward better operators.

5.5 Figures 6 and 7: Who rents in the land?

Read:

- Figure 6 shows that the increase in rented-in land is concentrated in the highest productivity quartile.



(a) By marginal revenue product **(b)** By total factor productivity (TFP)

FIGURE 3

Relationship between farm area and productivity before and after the reform

The figures show local polynomial regressions of household farm area on measures of household-level productivity. In panel (a), household productivity is the households marginal productivity averaged across crops and pre-reform years. In panel (b), household productivity is the households fixed TFP component from the production function regression in equation 4, averaged across crops.

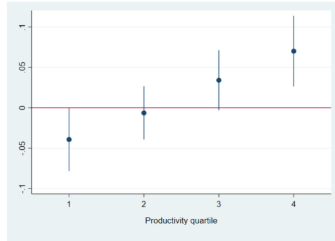
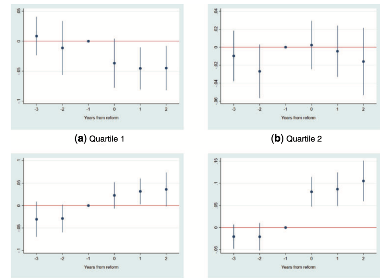


FIGURE 4

Reform effects on farm area, by productivity quartiles

The figure shows the coefficients and associated 90% confidence intervals from regressing the logarithm of crop-level area on productivity quartiles, where the quartiles are constructed from the village-crop level distribution of pre-reform marginal productivities.



(a) Quartile 1 **(b)** Quartile 2

(c) Quartile 3 **(d)** Quartile 4

FIGURE 5

Leads and lags of reform effects on farm area, by productivity quartiles

The panels in the figure plot the coefficients and associated 90% confidence intervals from estimating the leads and lags of reform effects separately for each of the marginal productivity quartiles, where the dependent variable is the logarithm of crop-level area. All effects are relative to the year prior to the reform ($k = -1$).

- Lower-productivity quartiles show small or statistically insignificant increases in renting in.
- Figure 7 confirms the timing: only the upper quartiles display a clear post-reform upward shift.
- Put differently, the reform does not just thicken the land market mechanically; it thickens it in an efficiency-enhancing direction.

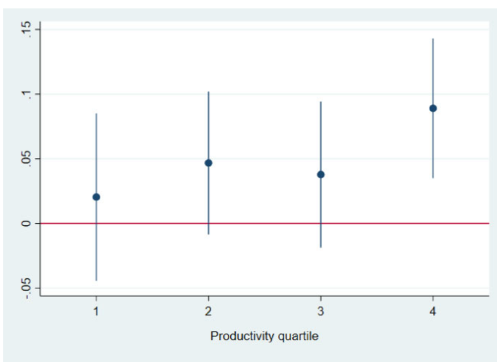
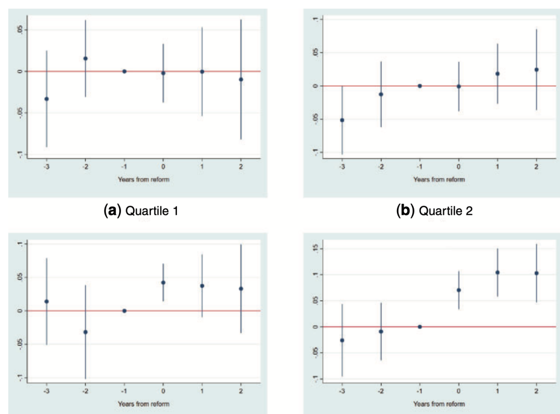


FIGURE 6

Reform effects on land rented in, by productivity quartiles

The figure shows the coefficients and associated 90% confidence intervals from regressing the IHS function of land rented in on productivity quartiles, where the quartiles are constructed from the village-level distribution of household-level fixed TFP.



(a) Quartile 1 **(b)** Quartile 2

(c) Quartile 3 **(d)** Quartile 4

FIGURE 7

Leads and lags of reform effects on area of land rented in, by productivity quartiles

The panels in the figure plot the coefficients and associated 90% confidence intervals from estimating the leads and lags of reform effects separately for each of the marginal productivity quartiles, where the dependent variable is the IHS function of the area of land rented in. All effects are relative to the year prior to the reform ($k = -1$).

5.6 Table 4: Capital and labor responses by productivity quartile

Read:

- There is no strong complementary response in agricultural capital or machinery cost. The reform is mainly about reallocating land, not about inducing a mechanization boom.
- Hired labor rises in the top productivity quartile: the post-reform quartile-4 coefficient is 0.0253, significant at the 10% level.
- Migration days and local off-farm days do not respond significantly. So labor reallocation happens mainly *within* village agriculture, not through large exits out of agriculture.
- This helps narrow the channel: the reform improves within-agriculture efficiency more than structural transformation.

TABLE 4

Impact of reform on capital and labour outcomes by productivity quartiles

	Agricultural capital (1)	Machine cost (2)	Hired labour days (3)	Migrant days (4)	Off-farm days (5)
Post-reform × Quartile 1	0.0519	−0.0195	0.0187	−0.0174	−0.0879
	(0.0927)	(0.0733)	(0.0129)	(0.0774)	(0.0870)
Post-reform × Quartile 2	0.0581	−0.0361	0.0194	−0.0568	−0.0494
	(0.0901)	(0.0813)	(0.0131)	(0.0819)	(0.0944)
Post-reform × Quartile 3	−0.0360	−0.00507	0.0202	−0.0186	−0.0438
	(0.0878)	(0.0804)	(0.0143)	(0.0670)	(0.0924)
Post-reform × Quartile 4	−0.100	0.0263	0.0253*	−0.00675	0.0249
	(0.0920)	(0.0840)	(0.0141)	(0.0510)	(0.0968)
Observations	113,209	275,540	275,540	113,209	113,209

Notes: In Columns 1, 4, and 5, each observation is a household-year. In Columns 2 and 3, each observation is a household-crop-year. The dependent variable is the IHS function of the column label. The quartiles refer to the village-level distribution of pre-reform marginal productivity. The regressions include household fixed effects, year fixed effects, the agricultural tax rate, the interaction between each marginal productivity quartile, and *ReformYear*. Columns 2 and 3 also include crop fixed effects. Standard errors clustered at the province level are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

TABLE 5

Impact of reform on components of aggregate TFP and aggregate TFP growth

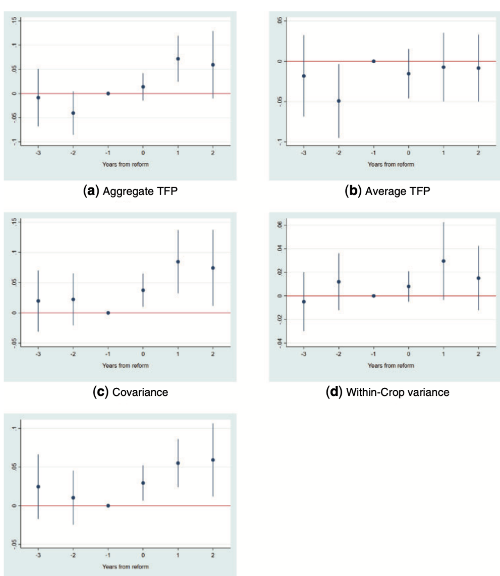
	Aggregate TFP					Aggregate TFP growth				
	Aggregate TFP (1)	Average TFP (2)	OP Cov (3)	Within Crop (4)	Between Crops (5)	Aggregate TFP growth (6)	Between Farm (7)	Within Farm (8)	Entry (9)	Exit (10)
Post-reform year	0.0755**	0.00888	0.0666**	0.0360*	0.0306	0.0567**	0.0441*	0.0118	0.00175	0.000967
	(0.0277)	(0.0203)	(0.0250)	(0.0202)	(0.0199)	(0.0263)	(0.0229)	(0.0197)	(0.00698)	(0.0170)
Reform year	0.0297	0.00105	0.0286	0.0175	0.0111	0.0245	0.0172	0.00133	0.00548	−0.000494
	(0.0208)	(0.0156)	(0.0180)	(0.0125)	(0.0149)	(0.0245)	(0.0178)	(0.0213)	(0.00764)	(0.0126)
Observations	2232	2232	2232	2232	2232	1964	1964	1964	1964	1964

Notes: The dependent variables in Columns 1–5 are the components of the Olley–Pakes decomposition of aggregate TFP. The dependent variables in Columns 6–10 are the components of the Baily–Hulten–Campbell decomposition of aggregate TFP growth. The unit of observation is a village-year. All regressions include village fixed effects, year fixed effects, and the agricultural tax rate. Standard errors clustered at province level are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

5.7 Table 5 and Figures 8–9: Decomposing the productivity gains

Read:

- Aggregate TFP rises by 0.0755, while average TFP rises by only 0.00888 and is not significant.
- The Olley–Pakes covariance term rises by 0.0666, meaning about 88% of the aggregate TFP gain comes from better reallocation, not from average technology improvements.
- The within-crop covariance component is positive and significant (0.0360); the between-crop component is similar in magnitude (0.0306) but less precise.
- In the BHC decomposition, aggregate TFP growth rises by 0.0567, mainly through the between-farm component (0.0441). Entry and exit contribute very little.
- Figures 8 and 9 again show that the changes occur after the reform, reinforcing the causal interpretation.

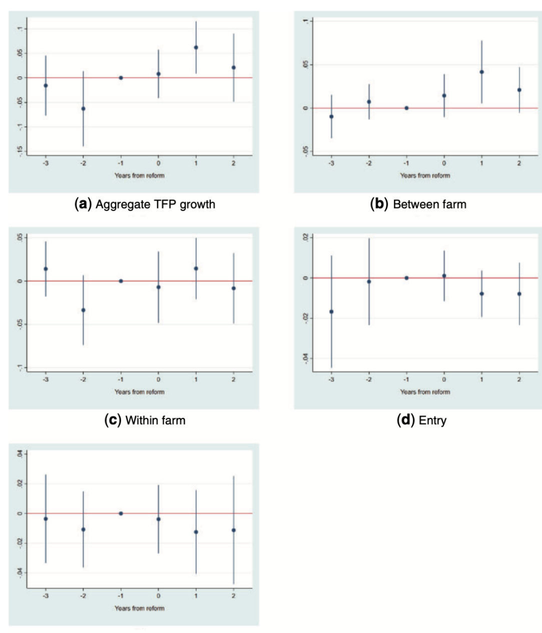


(a) Aggregate TFP **(b)** Average TFP **(c)** Covariance **(d)** Within-Crop variance **(e)** Between-crop variance

FIGURE 8

Leads and lags of reform effects on components of the Olley–Pakes decomposition

The panels in the figure plot the coefficients and associated 90% confidence intervals from estimating the leads and lags of reform effects, separately for each of the components of the Olley–Pakes decomposition of aggregate TFP. All effects are relative to the year prior to the reform ($k = -1$).



(a) Aggregate TFP growth **(b)** Between farm **(c)** Within farm **(d)** Entry **(e)** Exit

FIGURE 9

Leads and lags of reform effects on components of aggregate TFP growth

The panels in the figure plot the coefficients and associated 90% confidence intervals from estimating the leads and lags of reform effects, separately for each of the components of the Baily–Hulten–Campbell (BHC) decomposition of aggregate TFP growth. All effects are relative to the year prior to the reform ($k = -1$).

5.8 Tables 6 and 7: Responses to crop prices

Read:

- Table 6 validates the instrument: U.S. prices strongly predict Chinese cash-crop prices and only weakly predict staple-crop prices.
- In Table 7, the interaction between lagged crop price and post-reform status is positive in both reduced form and IV. For cash crops, a standard deviation increase in the price index raises cultivated area by about 2.7% more after reform.
- The probability that any land is allocated to a crop also rises after reform when that crop's price increases.
- The message is that better land markets improve not just the level of efficiency but also the economy's responsiveness to relative-price signals.

TABLE 6

Relationship between Chinese provincial and U.S. crop prices

	Cash crops (1)	Staple crops (2)
U.S. crop price	0.334***	0.0908**
	(0.0561)	(0.0359)
Observations	665	948

Notes: Each observation is a province-crop-year. The dependent variable is the crop price in a Chinese province. The regressions include fixed effects for crop, year, and province, and the agricultural tax rate. Standard errors clustered at the province level are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

TABLE 7

Impact of price and property rights reform on land allocation

	IHS area		I(any area)	
	Reduced form (1)	IV (2)	Reduced form (3)	IV (4)
Lagged price × Post-reform year	0.0602*	0.109*	0.0301**	0.0535**
	(0.0321)	(0.0550)	(0.0115)	(0.0200)
Lagged price × Reform year	0.0750**	0.230*	0.0267	0.0789
	(0.0294)	(0.113)	(0.0174)	(0.0600)
Post-reform year	−0.00184	−0.0104	0.00589	0.00173
	(0.0129)	(0.0116)	(0.00892)	(0.00848)
Reform year	0.00633	0.0234	0.00452	0.00987
	(0.00814)	(0.0139)	(0.00582)	(0.00797)
Observations	444,545	444,545	444,545	444,545

Notes: Each observation is a household-crop-year. The dependent variables are IHS area in Columns 1 and 2, and a dummy variable for any area planted in Columns 3 and 4. In Columns 1 and 3, lagged price is measured as U.S. crop price from the previous year. In Column 2 and 4, lagged price refers to the Chinese crop price from the previous year, instrumented with U.S. prices. The sample is restricted to cash crops only. The regressions include controls for agricultural taxes, household-crop fixed effects, and crop-year fixed effects. Standard errors clustered at the province level are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

6 Short summary

- **Conclusion.** Formalizing leasing rights in rural China increased land-market activity and raised agricultural output and land productivity.
- **Intuition.** Once leasing contracts became more secure, low-productivity farmers were more willing to lease out land and high-productivity farmers were better able to expand.
- **Empirical lesson.** Exchange rights matter a lot. Even without full privatization, reducing land-market frictions can generate large efficiency gains through reallocation.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** The RLCL caused a measurable increase in land rental activity and a sizable rise in village-level output and productivity.
- **Magnitude.** The reform raises land rented in by roughly 7% to 10%, village output by about 8%, and land productivity by about 10%.
- **Mechanism evidence.** Productive farmers expand, unproductive farmers contract, hired labor rises on the most productive farms, and the Olley–Pakes covariance term improves sharply.

- **Why this is credible.** The paper combines staggered implementation, fixed effects, event studies, adoption-timing tests, controls for agricultural taxes, land-quality robustness, and randomization inference.
- **Dynamic implication.** The reform also strengthens acreage responses to crop-price movements, so the gains are not only static.
- **Counterfactual implication.** Because this is only a partial reform, the large observed marginal effects imply that the full gains from eliminating land distortions could be much larger; the paper's benchmark is about 73%.

7.2 Economic intuition

- In a distorted land market, a farmer's operational scale reflects administrative assignment and contracting frictions rather than productive ability.
- Legal protection for leasing contracts lowers transaction costs and dispute risk, allowing land to move toward farmers with higher marginal products.
- Because land and labor are complementary, more productive farmers also hire more labor, but the reform does not appear strong enough to trigger a large reallocation of labor out of agriculture altogether.

7.3 Takeaway

This paper is important because it gives clean quasi-experimental evidence that property rights over *exchange* can matter enormously for allocative efficiency. The reform does not privatize land, yet it still generates large output gains because it lets land move toward more productive farmers and toward crops with better price signals. The broader lesson is that land institutions in developing countries should be evaluated not only for their effects on investment and security, but also for how they shape the reallocation of factors across producers.